

WORD PROBLEMS ACROSS THE STRICT–WEAK BOUNDARY IN CUBICAL TYPE THEORY

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ABSTRACT. Over a cellular context — the context of a cubical computad, freely listing cell variables over a De Morgan interval theory — two word problems coexist: whether two definable cells are equal *strictly* (by the decidable algorithmic equality of a cubical kernel) and whether they are equal *up to homotopy* (by a propositional identification). We show that the boundary between the strict and the weak layer is exactly the boundary between decidability and undecidability. The strict word problem is decidable over every cellular context, in every dimension. The propositional word problem is decidable over free graph contexts (dimension 1), where we make both halves of the decision procedure computational: identifications are compiled from generic groupoid laws, and separations are computed by a univalent transport invariant with a *proof-free carrier* — the faithful representation $F_2 \hookrightarrow \mathrm{SL}_2(\mathbb{Z})$ of Sanov, which evaluates at machine speed (under a millisecond per letter) where the reduced-word invariant is impractical. From dimension 2 on, the propositional problem is undecidable: finitely presented groups embed as relator cells, and both directions of the reduction are exhibited inside the kernel — relator applications compile to explicit path constructions over the generic presentation context, and the presentation complex of $\mathbb{Z}/2$ is realized internally as $\mathbb{RP}^2$, where a univalent double cover certifies that the presentation is realized faithfully ($x^2 = 1$ and $x \neq 1$, both machine-checked). All positive claims are anchored in a kernel-checked artifact; the measured computational frontier of the kernel itself is reported.

1. INTRODUCTION

A cubical proof assistant maintains two equalities. The *strict* (definitional, algorithmic) equality $\equiv_{\mathrm{alg}}$ is the one its kernel decides silently at every type check; the *weak* (propositional) equality $\equiv_{\mathrm{prop}}$ is the one its users prove, path by path. A companion line of work [6, 7, 8, 9] has measured the gap between the two layers over *cellular contexts* — contexts Γ that freely list cell variables ($A : \mathcal{U}$, points, paths, squares, $\dots$), the type-theoretic incarnation of cubical computads — and found the strict layer to be small, combinatorial, and well-organized: strict cells are classified by

2020 *Mathematics Subject Classification.* 03D40, 03B38, 18N45, 20F10.

Key words and phrases. cubical type theory, word problem, decidability, cubical computads, covering spaces, univalence, free groups, group presentations.

reparametrizations along the free De Morgan algebra, the strict layer of a generic n -cell is a wedge of De Morgan spheres, cellular contexts realize their cube presheaves on the nose, and the strict fibers over a boundary are strictly contractible.

This paper asks the decision-theoretic question: *given two definable cells over a cellular context, can a machine decide whether they are equal?* The answer splits along the same seam:

	dimension 1 (free)	dimension ≥ 2 (with relators)
strict $\equiv_{\text{alg}}$	decidable	decidable
weak $\equiv_{\text{prop}}$	decidable	undecidable

The strict word problem is decidable over *every* cellular context (Theorem 3.1): the kernel’s algorithmic equality is a decision procedure, and the realization theorem of [8] says exactly that it decides the intended combinatorial equality. The propositional word problem is decidable in dimension 1 (Theorem 4.1) and undecidable from dimension 2 on (Theorem 9.1): group presentations embed as relator cells and the word problem of a finitely presented group reduces to $\equiv_{\text{prop}}$, so Novikov–Boone applies [4, 1]. *The strict–weak boundary is the decidable–undecidable boundary.* This gives a quantitative reading of what weakening buys and what it costs: relator cells create the identifications that make homotopy theory expressive, and those same identifications destroy decidability — while the strict layer, which by the no-go theorems of [6, 9] can never absorb a relator, remains decidable throughout.

The mathematical content of the boundary statement is classical: normal forms for free groupoids on one side, Novikov–Boone on the other. The contribution of this paper is to make the statement *computational* in a live cubical kernel, in four steps.

- (1) *Identifications compile.* Over the free context, the identification half of the dimension-1 decision procedure is a compiler from reduction sequences of words to explicit path composites of generic groupoid laws, each machine-checked once and for all (§4). Over a presentation context, relator applications compile the same way: the group-theoretic consequence $x^4 = 1$ of the single relation $x^2 = 1$ becomes an explicit path constructed from two relator uses and one unit law, checked over the *generic* presentation context (§6).
- (2) *Separations compute.* The separation half needs invariants that evaluate. Abelian invariants (winding numbers) are cheap but incomplete. We introduce a transport invariant with a *proof-free carrier*: the faithful Sanov representation $F_2 \hookrightarrow \text{SL}_2(\mathbb{Z})$, implemented as a univalent cover of the figure eight whose fibers are $\mathbb{Z}^4$ and whose monodromies are elementary integer-arithmetic equivalences. Replacing a proof-carrying carrier (reduced words as a subset type) by a flat one moves the per-letter cost from “infeasible” to under

a millisecond, and faithfulness makes the invariant complete for F_2 (§5).

- (3) *Presentations realize.* For the undecidability transfer, the presentation complex is built *inside* the theory: for $\langle x \mid x^2 \rangle$ it is the mapping cone of the degree-2 map, i.e. $\mathbb{RP}^2$, and the relator cell folds into a machine-checked identification $x \cdot x \equiv_{\text{prop}} \text{refl}$ — one **hcomp** square capped at the cone point. Generic relator derivations then transfer by instantiation alone (§7).
- (4) *Faithfulness certifies.* A univalent double cover of $\mathbb{RP}^2$, whose existence hinges on exactly the coherence $\text{ua}(\text{not}) \cdot \text{ua}(\text{not}) \equiv_{\text{prop}} \text{refl}$, computes a $\mathbb{Z}/2$ winding and refutes $x \equiv_{\text{prop}} \text{refl}$. Together with $x \cdot x \equiv_{\text{prop}} \text{refl}$ this machine-certifies that the internal realization presents $\mathbb{Z}/2$ faithfully: $x^2 = 1$ and $x \neq 1$ (§8).

Along the way we report the measured computational frontier of the kernel itself (§11): transport along a single **hcomp**-composite of universe paths completes in about a minute, while three nested layers blow up superlinearly — the identified obstruction, value-sharing across Kan steps, is a kernel-engineering problem rather than a mathematical one, and the iterated per-letter procedure sidesteps it entirely.

Relation to the companion papers. The kernel, the constancy rule studied in [6], and the cellular-context platform of [7, 8, 9] are reused here; this paper is self-contained in its statements and cites the companions for proofs of the imported facts (§2). The artifact conventions follow the series: every machine-checked claim names a definition in the accompanying repository (§13).

2. SETTING AND IMPORTED FACTS

We work in the object theory of a defensively small cubical proof kernel implementing a CCHM-style cubical type theory [2]: dependent products and sums, universes $\mathcal{U}_0 : \mathcal{U}_1$ (no type-in-type), a De Morgan interval $\mathbb{I}$ with connections and reversal, **Path/PathP** types, **transp/hcomp** with the standard computation rules, Glue types with univalence (**ua**) computing by unglueing, and the higher inductive types used below (S^1 , pushouts). Its algorithmic equality $\equiv_{\text{alg}}$ is a terminating normalization-by-evaluation conversion check; *decidability of $\equiv_{\text{alg}}$ on well-typed terms* is the kernel’s basic metatheoretic budget, and everything labeled “machine-checked” below is a type check (positive or, for separations, deliberately failing) or a closed-term normalization performed by this kernel.

Definition 2.1 (cellular context). A *cellular context* is a context of the object theory of the shape

$$\Gamma = A : \mathcal{U}, a_1 : A, \dots, p_1 : \text{Path } A a_i a_j, \dots, Q_1 : \text{Sq}(\dots), \dots$$

listing finitely many cell variables whose boundaries are formed from earlier cells — the context of a finite cubical computad (polygraph). A *definable*

cell over Γ is a term of a cell type over Γ ; a *word* is a definable cell built from cell variables by path operations (*refl*, *inversion*, *cong*, *composition* $\cdot$, and *Kan fillers*).

We import three facts from the companions, with the anchor statements re-proved in the artifact of this paper where they are used.

- (K1) *Strict classification* [7]: over a generic cellular context, the strict inhabitants of a cell type are exactly the reparametrizations of cell variables along the free De Morgan algebra on the bound interval variables; the strict layer of a generic n -cell is the wedge-of-spheres monoid $\text{DM}(-)$ -set.
- (K2) *Realization* [8]: cellular contexts realize their cube presheaves on the nose — the attachment equations of the computad hold definitionally, and the kernel's $\equiv_{\text{alg}}$ neither merges distinct presheaf cells nor misses an identification.
- (K3) *Decidability of the strict layer* [7, 8]: equality of reparametrizations in the free De Morgan algebra is decidable (disjunctive normal forms of antichains), and $\equiv_{\text{alg}}$ restricted to strict cells implements it.

Two further companion results frame the interpretation: the no-go theorems of [6, 9] (a nontrivial relator can never be made definitional — constant-tube *hcomp*-regularity at path types breaks canonicity-adjacent invariants), and the strict contractibility of filler fibers [9]. They will reappear in §10 as the conceptual reason the decidability boundary sits where it does.

3. THE STRICT WORD PROBLEM IS DECIDABLE EVERYWHERE

Theorem 3.1. *Over every cellular context Γ and in every dimension, the strict word problem — given two definable cells u, v of a common cell type over Γ , decide $u \equiv_{\text{alg}} v$ — is decidable. Moreover the decision agrees with equality in the cube presheaf presented by Γ : relator cells are treated as generating cells, never as equations.*

Proof. Decidability is the kernel's conversion check on well-typed terms (§2). Agreement with the presheaf semantics is (K2): the realization theorem states that $\equiv_{\text{alg}}$ over Γ decides exactly the on-the-nose equality of the presented presheaf, with the strict layer classified by (K1) and its equational theory decided by (K3). A relator cell — a cell variable whose boundary happens to spell a word — contributes generators to the presheaf, not equations: no computation rule of the kernel consults the boundary of a variable to rewrite a term. (This is the operational face of the no-go theorems: the kernel *could not* consistently promote the relator to a rewrite even if asked; cf. §10.) $\square$

The theorem is deliberately unspectacular — it records that the strict layer stays decidable *uniformly*, including over the very contexts on which the propositional layer will turn undecidable. That uniformity is the left column of the table in §1, and half of the boundary statement.

4. DIMENSION ONE, FREE: THE PROPOSITIONAL PROBLEM IS DECIDABLE

Let Γ_G be a free graph context: a type A , finitely many points, and finitely many path variables (edges), with no cells of dimension ≥ 2 . Words over Γ_G are edge-words: composites of edges and their inverses. Classically, the free groupoid on a graph has normal forms — reduced words — and two edge-words are equal in the free groupoid iff their reductions coincide. The propositional layer of the kernel presents exactly the free groupoid:

Theorem 4.1. *Over a free graph context, the propositional word problem is decidable: $w_1 \equiv_{\text{prop}} w_2$ is inhabited iff the reduced words of w_1 and w_2 coincide. Both halves of the decision procedure are computational over the kernel:*

- (1) (identification) *if the reductions coincide, an inhabitant of $w_1 \equiv_{\text{prop}} w_2$ is compiled from the generic groupoid laws — associativity, units, inverses — each a machine-checked path over the generic context;*
- (2) (separation) *if the reductions differ, the words are separated by instantiating Γ_G at a univalent cover and computing a transport invariant; since substitution preserves $\equiv_{\text{prop}}$, distinct invariant values over the instance refute $w_1 \equiv_{\text{prop}} w_2$ over the generic context.*

Proof scheme. Completeness of reduced words for the free groupoid is classical. (1) is a compilation: each reduction step (cancel $e \cdot e^{-1}$, reassociate, absorb a unit) is an instance of a generic law available as a closed library path (`transAssoc`, `transReflL/R`, `transInvR/L` in the artifact), transported into word position by `cong` and chained by `;`; the compiled composite type-checks over Γ_G itself. (2) reduces to exhibiting, for each pair of distinct reduced words, an instance and an invariant with distinct computed values. For a single edge instantiated at the circle, the winding invariant computes $(+1, -1, 0, +2, \dots)$ (are among the artifact’s normalized values); for several edges the wedge of circles is the universal instance, and §5 supplies a complete, practically computable invariant. Soundness of the refutation is transport invariance: an identification over Γ_G would transport to an identification of distinct normal forms of the invariant’s values. $\square$

The interesting engineering content is in (2): completeness of the separation half needs an invariant that (a) distinguishes all pairs of distinct reduced words — so abelian invariants are insufficient — and (b) actually evaluates on the kernel. The next section constructs one.

5. A PROOF-FREE CARRIER: THE SANOV INVARIANT

The natural complete invariant for two generators is the F_2 -cover of the figure eight $S^1 \vee S^1$: fibers are reduced words, monodromy prepends a generator with cancellation. The series artifact contains this cover in full, machine-checked — and it is computationally impractical. Its carrier is a subset type (words with a reducedness proof), so every value carries a

proof, and the cover equivalence embeds a proof that the carrier is a set; the kernel’s Kan steps traverse those proofs. A single generator’s winding normalizes in seconds; composites exhaust any practical budget. The failure mode is structural, not accidental: *proof-carrying carriers make transport walk proofs*.

The remedy is to change the invariant, not the kernel. Sanov’s representation [5]

$$L \mapsto \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad R \mapsto \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

embeds F_2 faithfully into $\mathrm{SL}_2(\mathbb{Z})$. The carrier is now $\mathbb{Z}^4$ — a matrix, a pair of pairs of integers, with *no propositional payload*.

Definition 5.1 (Sanov cover). Right multiplication by L (resp. R) is an equivalence of $\mathbb{Z}^4$: the inverse prepends the negated double, and both round trips reduce to two integer cancellation laws ($x + (-x + y) = y$ and its mirror), proved from associativity, inverses and units of $\mathbb{Z}$. The *Sanov cover* $\mathrm{helixSL} : S^1 \vee S^1 \rightarrow \mathcal{U}$ sends each circle to **ua** of its generator’s equivalence; the *matrix winding* windSL of a loop transports the identity matrix along the cover.

Theorem 5.2. *The Sanov invariant is complete for the free group on two generators and computes at machine speed on the kernel:*

- (1) *generators wind to their Sanov matrices in milliseconds (measured: 8 ms, against seconds for the reduced-word cover);*
- (2) *the iterated procedure — one Glue transport per letter, feeding the concrete matrix forward — runs at under a millisecond per letter; in particular $L \cdot R \mapsto \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}$ and $R \cdot L \mapsto \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$ are distinct — separating a pair that every abelian invariant conflates — and the commutator $L \cdot R \cdot L^{-1} \cdot R^{-1}$ evaluates to $\begin{pmatrix} 21 & -8 \\ 8 & -3 \end{pmatrix} \neq I$;*
- (3) *two words are equal in F_2 iff their computed matrices coincide (faithfulness of the representation), so the invariant decides the separation half of Theorem 4.1 completely, in time linear in word length.*

*The agreement of the iterated procedure with the direct winding of the composite loop is propositional and generic: **cong** distributes over composition (**congTrans**), and transport along a composite of universe paths is the composite of transports (**transpTrans**), both closed library paths.*

Proof. (1) and (2) are kernel computations, machine-checked against the expected matrices in the artifact’s native harness; (3) is Sanov’s theorem [5] together with soundness of the computed values (canonicity for the closed integer type: a normalized matrix entry is a numeral, and distinct numerals are separated by the decidable equality of $\mathbb{Z}$, itself a library fact). The bridging identifications are the named library paths. $\square$

Remark 5.3. The lesson we draw — and use again in §8 — is a design principle for computational univalence: *represent invariants on proof-free*

carriers, pushing all proof obligations into the equivalences. The equivalence proofs are walked once per Glue formation, not once per transported value; normal forms of values stay small; and a classical faithful representation (here into $\mathrm{SL}_2(\mathbb{Z})$) converts completeness of the invariant into a theorem about the target group rather than about the type-theoretic carrier.

6. RELATORS COMPILE: THE PRESENTATION CONTEXT

From dimension 2 on, cellular contexts contain *relator cells*: squares whose boundary spells a word. Up to the standard globular–cubical translation, the presentation context of $\langle x \mid x^2 \rangle$ is

$$\Gamma_{\mathbb{Z}/2} = A : \mathcal{U}, a : A, x : \mathrm{Path} \, A \, a \, a, R : \mathrm{Path} \, (\mathrm{Path} \, A \, a \, a) \, (x \cdot x) \, \mathrm{refl}.$$

Over $\Gamma_{\mathbb{Z}/2}$, the group-theoretic consequences of the relation compile to explicit paths, exactly as the groupoid laws did in Theorem 4.1(1) — with R now among the usable moves:

Proposition 6.1. *Over $\Gamma_{\mathbb{Z}/2}$ there is a machine-checked path*

$$\begin{aligned} \mathrm{wordX4} & : (x \cdot x) \cdot (x \cdot x) \equiv_{\mathrm{prop}} \mathrm{refl}, \\ \mathrm{wordX4} & = \mathrm{cong}_{(-) \cdot (x \cdot x)}(R) \cdot \mathrm{transRefL} \cdot R, \end{aligned}$$

*i.e. $x^4 = 1$ is derived by two relator uses and one unit law. The compilation scheme is uniform: a derivation of $w = 1$ in the presented group maps to a path $w \equiv_{\mathrm{prop}} \mathrm{refl}$ by sending each relator application to a **cong**-embedded use of the relator variable and each free-reduction step to a groupoid law, chained by $\cdot$.*

This is one half of the undecidability transfer: *provable group equalities become inhabitants*. It is worth noting what the compilation does *not* need: no instance, no Kan machinery beyond composition, no knowledge of the ambient type A — the derivation lives over the generic context and transfers to every model by substitution. The other half — *inhabitants imply group equalities* — needs an interpretation, hence a realization of the presentation complex; that is the next section.

7. PRESENTATIONS REALIZE: $\mathbb{RP}^2$ INSIDE THE KERNEL

The presentation complex of $\langle x \mid x^2 \rangle$ — one vertex, one edge, one 2-cell attached along x^2 — is the mapping cone of the degree-2 self-map of the circle: the real projective plane. Both the attaching map and the cone exist as first-class objects of the kernel:

Definition 7.1. $\mathrm{deg}_2 : S^1 \rightarrow S^1$ is the circle eliminator with $\mathrm{base} \mapsto \mathrm{base}$ and $\mathrm{loop} \mapsto \mathrm{loop} \cdot \mathrm{loop}$; $\mathbb{RP}^2 := \mathrm{cofib}(\mathrm{deg}_2)$ is its mapping cone, the pushout $1 \leftarrow S^1 \xrightarrow{\mathrm{deg}_2} S^1$. Write $b := \mathrm{inr}(\mathrm{base})$ and $x := \langle i \rangle \mathrm{inr}(\mathrm{loop} \, i)$ for the realized vertex and generator.

Theorem 7.2. *The internal $\mathbb{RP}^2$ realizes the presentation:*

- (1) (attachment, strictly) $\text{cong}_{\deg_2}(\text{loop}) \equiv_{\text{alg}} \text{loop} \cdot \text{loop}$ — the attaching map computes definitionally on the generator — and the cone path from the cone point to b is a definable cell;
- (2) (the relator cell, realized) there is a machine-checked identification $\text{rp2Rel} : x \cdot x \equiv_{\text{prop}} \text{refl}_b$, obtained by folding the pushout cylinder with a single hcomp square capped at the cone point, then splitting the image of $\text{loop} \cdot \text{loop}$ by congTrans ;
- (3) (transfer by instantiation) the substitution of $(\mathbb{RP}^2, b, x, \text{rp2Rel})$ for (A, a, x, R) in Proposition 6.1 yields a machine-checked identification $x^4 \equiv_{\text{prop}} \text{refl}$ in $\mathbb{RP}^2$: generic relator derivations land on the internal realization with no further work;
- (4) (properly propositional) the identification in (2) is not strict: the constant path-lambda fails to type-check against $x \cdot x \equiv_{\text{prop}} \text{refl}$ (a deliberate negative probe), as the no-go theorems require.

Proof. All four items are single artifact probes. (1) the constant path-lambda inhabits $\langle i \rangle \deg_2(\text{loop } i) \equiv_{\text{prop}} \text{loop} \cdot \text{loop}$, so the attachment is definitional; (2) the square has the pushout cylinder as its $k=0$ wall, the cone point as cap, and the basepoint cone path on the remaining three walls; (3) is literally an application of the closed term `wordX4`; (4) is a failing type check, recorded as a guard. $\square$

8. FAITHFULNESS: THE DOUBLE COVER OF $\mathbb{RP}^2$

Theorem 7.2 would be vacuous if the realization collapsed — if, say, $x \equiv_{\text{prop}} \text{refl}$ held in $\mathbb{RP}^2$, every word would be identified and the relator would certify nothing. The classical statement $\pi_1(\mathbb{RP}^2) = \mathbb{Z}/2$ says precisely that the presentation is realized *faithfully*: $x^2 = 1$ (the relator) and $x \neq 1$ (no collapse). We certify the computational core of this on the kernel with a univalent double cover, built by the proof-free-carrier principle of §5 with carrier `Bool`.

Lemma 8.1. $\text{ua}(\text{not}) \cdot \text{ua}(\text{not}) \equiv_{\text{prop}} \text{refl}_{\text{Bool}}$, machine-checked (by injectivity of ua , equality of equivalences from equality of functions, and the pointwise involution $\text{not} \circ \text{not} = \text{id}$).

Theorem 8.2. There is a cover $\text{covRP2} : \mathbb{RP}^2 \rightarrow \mathcal{U}$ with fiber `Bool`, sending the 1-skeleton circle to $\text{ua}(\text{not})$; its extension over the relator 2-cell requires exactly Lemma 8.1, and is constructed by transporting the trivial square along it (path induction in $\mathcal{U}_1$). The resulting winding $\text{windRP2} := \text{transp}_{\text{covRP2}}(\text{true})$ computes:

$$\text{windRP2}(\text{refl}) \mapsto \text{true}, \quad \text{windRP2}(x) \mapsto \text{false},$$

whence a machine-checked refutation

$$\text{rp2LoopNontriv} : (x \equiv_{\text{prop}} \text{refl}) \rightarrow \mathbf{0}.$$

Together with **rp2Rel** (Theorem 7.2(2)), the presentation $\langle x \mid x^2 \rangle$ is realized faithfully at the level of loop identifications: $x^2 = 1$ and $x \neq 1$. As a corollary, $\mathbb{RP}^2$ is not a set.

Proof. The cover is a pushout eliminator into $\mathcal{U}$; on the cone it is constantly **Bool**, on the 1-skeleton it is the circle eliminator with loop payload $\mathbf{ua}(\mathbf{not})$, and on the pushout cylinder it must provide, over each point of the attaching circle, a path $\mathbf{Bool} \equiv_{\text{prop}} \mathbf{cov}(\deg_2(-))$ — a square whose monodromy side is the image of $\mathbf{loop} \cdot \mathbf{loop}$, i.e. (after **congTrans**) the composite $\mathbf{ua}(\mathbf{not}) \cdot \mathbf{ua}(\mathbf{not})$. Lemma 8.1 is thus literally the extension condition; path induction along it reduces the required square to the trivial one. The two windings are closed-term normalizations; the refutation composes **cong_{windRP2}** with the decidable separation of **false** from **true**. $\square$

Remark 8.3. The cover makes a second architectural point: the extension condition over a relator cell is always a coherence in the universe — here $\mathbf{ua}(\mathbf{not})^2 \equiv_{\text{prop}} \mathbf{refl}$, in general the image of the relator word under the intended monodromy. Constructing covers of internal presentation complexes is thus exactly as hard as proving the corresponding relation among **ua**-paths, a task the proof-free-carrier principle keeps tractable.

9. FROM DIMENSION TWO: UNDECIDABILITY

Theorem 9.1. *The propositional word problem over cellular contexts with cells of dimension ≥ 2 is undecidable: there is no algorithm that, given a cellular context Γ and two edge-words w_1, w_2 over it, decides whether $w_1 \equiv_{\text{prop}} w_2$ is inhabited.*

Proof, by reduction. Let $\langle X \mid \mathcal{R} \rangle$ be a finite group presentation and $\Gamma_{\langle X \mid \mathcal{R} \rangle}$ its presentation context: one point, one loop per generator, one relator square per relation (as in §6). We claim $w \equiv_{\text{prop}} \mathbf{refl}$ is inhabited over $\Gamma_{\langle X \mid \mathcal{R} \rangle}$ iff $w = 1$ in the presented group G ; undecidability then follows from the existence of finitely presented groups with undecidable word problem [4, 1].

$(\Leftarrow)$ is the compilation of Proposition 6.1, verbatim for arbitrary presentations. $(\Rightarrow)$: interpret $\Gamma_{\langle X \mid \mathcal{R} \rangle}$ into the realization of its presentation complex $K_{\langle X \mid \mathcal{R} \rangle}$ — the interpretation exists in any model of the theory with pushouts, sending cell variables to the corresponding cells, as in Theorem 7.2 for the one-relator case — and apply $\pi_1(K_{\langle X \mid \mathcal{R} \rangle}) = G$ (van Kampen): an inhabitant of $w \equiv_{\text{prop}} \mathbf{refl}$ over the generic context substitutes to an identification in the realization, hence to $w = 1$ in G . Soundness of the type theory for such models is the standard cubical-sets semantics [2]. $\square$

Remark 9.2. The kernel-checked material of §§6–8 exhibits both directions of the reduction in the minimal nontrivial instance $\mathbb{Z}/2$: the compilation $(\Leftarrow)$ as **wordX4**, and the faithful realization underlying $(\Rightarrow)$ as **rp2Rel** + **rp2LoopNontriv** — including the genuinely propositional character of the relator (Theorem 7.2(4)). We do not claim an internal proof of $\pi_1 = G$ for

arbitrary presentations; the reduction above uses it classically, as is standard for undecidability transfers.

10. THE BOUNDARY

Assembling Theorems 3.1, 4.1 and 9.1:

	dimension 1 (free)	dimension ≥ 2 (relators)
strict $\equiv_{\text{alg}}$	decidable (K1–K3)	decidable (Theorem 3.1)
weak $\equiv_{\text{prop}}$	decidable (Theorem 4.1)	undecidable (Theorem 9.1)

The two layers separate exactly at the relator cells, and the no-go theorems of the series say this is not an accident of implementation. A kernel that tried to decide the propositional layer by promoting relators to definitional equations would need $R : x \cdot x \equiv_{\text{prop}} \text{refl}$ to hold strictly; the no-go theorems [6, 9] show that even the humblest such promotion — constant-tube **hcomp**-regularity at path types — already destroys the invariants behind canonicity. Undecidability gives the quantitative version: *no* extension of the strict equality can absorb the relators of an arbitrary presentation while remaining decidable, since the resulting theory would decide group word problems. Conversely, the price is worth paying exactly where it is paid: the weak layer’s identifications are what make the presentation complexes, covers and invariants of §§7–8 exist at all. The strict–weak boundary is not a defect line but a load-bearing wall: on one side, a decidable combinatorics of reparametrizations (K1–K3); on the other, the full undecidable expressiveness of homotopy theory — with the compilation of Proposition 6.1 and the covers of §8 as the traffic across it.

11. THE MEASURED FRONTIER

Two performance walls were identified and measured in the course of this work; both concern the kernel, not the mathematics, and both are reported here as data for kernel engineering.

Proof-carrying carriers (§5): transport walks the proofs embedded in carrier values and cover equivalences. With the reduced-word carrier a single generator costs seconds and composites are infeasible; the proof-free Sanov carrier removes this wall entirely (milliseconds per letter, measured).

Nested Kan composition in the universe: transporting directly along a composite loop evaluates **hcomp** in $\mathcal{U}$ over Glue lines. Measured on the Sanov cover: one nested layer (the cancelling composite, or the wedge-conjugated second generator) completes in about a minute; three nested layers exceed 50 CPU minutes without completing. The growth is superlinear in nesting depth; the identified cause is re-evaluation of inner Glue layers at each Kan step, and the identified remedy — value-sharing (memoization) across Kan steps — is a kernel-engineering project orthogonal to this paper. The iterated per-letter procedure of Theorem 5.2 avoids the wall by construction, and its agreement with the direct composite is the generic propositional

bridge (`congTrans`, `transpTrans`); nothing in Theorems 4.1–9.1 depends on the wall.

12. OUTLOOK

Three continuations suggest themselves. First, *internal π_1 for general presentations*: the $\mathbb{Z}/2$ case of §§7–8 used no property of the presentation beyond its finiteness; an internal encode–decode proof of $\pi_1(K_{\langle X|\mathcal{R} \rangle}) = G$ (in the style of [3], with the cover extension conditions discharged relator-by-relator as in Lemma 8.1) would make the ($\Rightarrow$) direction of Theorem 9.1 internal as well. Second, *up-to-homotopy normal forms*: over free contexts, reduced words are normal forms for the weak layer; whether a useful normal-form calculus survives some relator classes (one-relator groups, small cancellation) is the type-theoretic shadow of classical combinatorial group theory, with the compiled identifications of Proposition 6.1 as the rewriting moves. Third, *the kernel frontier*: value-sharing across Kan steps (§11) would make direct composite transport practical and is the single most valuable kernel improvement identified by this series.

Acknowledgements. The platform and companion results were developed in [6, 7, 8, 9].

13. ARTIFACT

All machine-checked claims are definitions with positive or negative guards in the accompanying repository (kernel, library and probe suites; the native harness reports the timings quoted above). Anchors: the presentation-context compilation (§6) is `wordX4/wordRel`; the Sanov invariant (§5) is `addCancel/addCancelN`, `sanovL/R`, `helixSL`, `windSL`, with the timing harness `Test/Sanov.lean`; the internal presentation complex (§7) is `deg2`, `deg2Loop`, `rp2`, `rp2Attach`, `rp2RelSq`, `rp2Rel`, `wordX4RP2`, `wordRelRP2`; the double cover (§8) is `uaNotNot`, `covS1`, `covRP2`, `windRP2`, `rp2LoopNontriv`, `rp2NotSet`; the groupoid laws and winding values of Theorem 4.1 and the reduced-word F_2 cover are in the library of the earlier releases. Repository: <https://github.com/r-shibusawa/cubical-strict-j>, archived at DOI 10.5281/zenodo.21405961 (concept DOI; the release accompanying this paper is tagged in the repository).

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